Practice workbook · Explain how diaphragm and intercostal muscles drive breathing

Muscle contraction changing thorax volume and pressure to move air, and what stops it.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VENTILATION-MUSCLE-ACTION

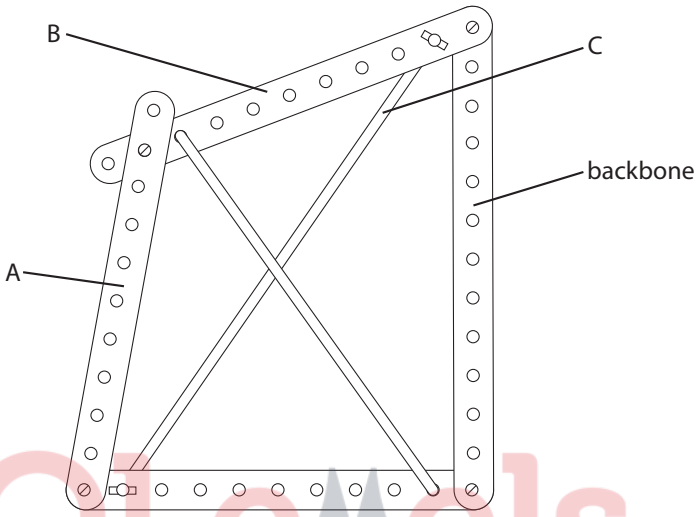
1. Question 5 · part a.iii · 6 marks

June 2025 Paper 1

Worked-example retry · Attempt part a.iii only.

5 (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.



(i) State which parts of the breathing system are represented by A, B and C.

(3)

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

(3)

1 .....

2 .....

3 .....

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P 7 8 7 6 8 A 0 1 2 2 0

1. Question 5 · part a.iii · 6 marks

June 2025 Paper 1

Worked-example retry · Attempt part a.iii only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

(6)

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

(3)

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(Total for Question 5 = 15 marks)

1. Question 5 · part a.iii · 6 marks · Working and answer

Working and answer

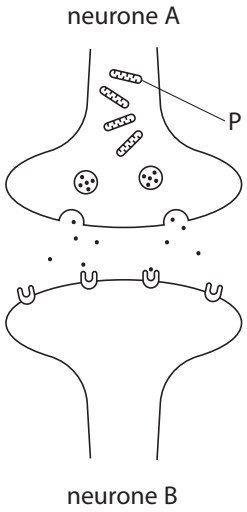
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2. Question 8 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse.

(1)

(ii) Explain the function of structure P in the transmission of the impulse across the gap.

(3)

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2. Question 8 · part b.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

(2)

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

(2)

(Total for Question 8 = 8 marks)

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2. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

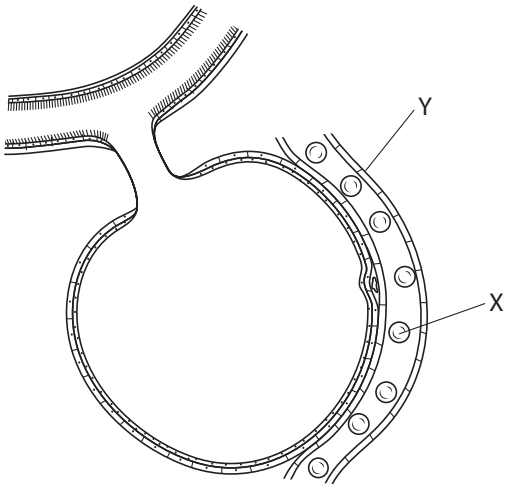
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3. Question 3 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

3 The diagram shows an alveolus and its blood supply.



(a) (i) What is the name of blood component X?

(1)

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere.

(3)

1 .....

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2 .....

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3 .....

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3. Question 3 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(iii) Give three features of alveoli that allow efficient gas exchange.

(3)

1 .....

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2 .....

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3 .....

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(b) Structure Y is a capillary.

Give two features in the diagram which show that structure Y is a capillary.

(2)

1 .....

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2 .....

.....

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(c) Some people have a condition known as emphysema. One symptom of emphysema is the breakdown of elastic fibres in the lung tissue.

Suggest the effects that loss of elastic fibres have on a person's ability to breathe.

(2)

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(Total for Question 3 = 11 marks)

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3. Question 3 · part c · 2 marks · Working and answer

Working and answer

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4. Question 7 · part a · 4 marks

June 2024 Paper 1

Further practice · Attempt part a only.

7 (a) The list gives some structures found in the thorax.

- intercostal muscles
- ribs
- diaphragm

Describe how these structures work together during inhalation.

(4)

(b) Describe the role of chemoreceptors in the carotid arteries.

(2)

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4. Question 7 · part a · 4 marks · Working and answer

Working and answer

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